HARMONIC COARSE LEFT INVERSES FOR QUASI-ISOMETRIC EMBEDDINGS OF THE HYPERBOLIC PLANE

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ABSTRACT. Let M be a pinched Hadamard manifold and $\iota\colon \mathbb{H}^2 \to M$ a quasi-isometric embedding. We prove that there is a harmonic map $h\colon M \to \mathbb{H}^2$ such that $h \circ \iota$ stays at bounded distance from $\mathrm{id}_{\mathbb{H}^2}$, answering a conjecture of Tošić.

1. INTRODUCTION

The relationship between harmonic maps and large scale geometry is an important theme in the study of negatively curved manifolds. Quasi-isometric embeddings capture the large scale geometry of metric spaces, while harmonic maps reflect their differential geometric structure. A natural question is whether a quasi-isometric embedding can be approximated, within uniformly bounded distance, by a harmonic map. Benoist and Hulin answered this question affirmatively for quasi-isometric embeddings between pinched Hadamard manifolds, namely, complete, simply connected Riemannian manifolds whose sectional curvatures are bounded between two strictly negative constants [1, Theorem 1.1].

A related question is whether a quasi-isometric embedding of the hyperbolic plane into a negatively curved manifold admits a harmonic map from the ambient manifold back to the hyperbolic plane that serves as an approximate left inverse.

To formulate this question precisely, let M be a pinched Hadamard manifold of dimension $n \geq 2$, whose sectional curvatures satisfy

$$-b^2 \leq \sec_M \leq -a^2 < 0 \tag{1}$$

for constants $0 < a \leq b$. We denote its Riemannian distance by d_M and write $\mathbb{H}^m$ for real hyperbolic m-space of constant sectional curvature -1. In this setting, Tošić formulated the following conjecture [12, Conjecture 1.2].

Conjecture 1.1 (Tošić)**.** *Let M be a pinched Hadamard manifold. For every quasi-isometric embedding $\iota\colon \mathbb{H}^2 \to M$, there exists a harmonic map $h\colon M \to \mathbb{H}^2$ such that*

$$\sup_{x \in \mathbb{H}^2} d_{\mathbb{H}^2}\big(h(\iota(x)), x\big) < \infty. \tag{2}$$

Tošić proved the conjecture when $M = \mathbb{H}^n$ [12, Theorem 1.3]. His approach begins with nearest point projection onto the convex hull of the limit set of $\iota(\mathbb{H}^2)$, followed by a coarse inverse defined on a neighborhood of the hull [12, Section 6.3]. Since the resulting map need not be a quasi-isometric embedding, the Benoist–Hulin

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theorem does not directly apply. Related results on harmonic approximation of convex hull projections [12, Theorems 1.4 and 1.5] require bounds on the invariant upper Minkowski dimension of the relevant subset of the ideal boundary and, in the general pinched Hadamard setting, a coboundedness assumption on the isometry group action.

Our main result establishes this conjecture.

Theorem 1.2. *Conjecture 1.1 holds.*

The main analytic ingredient is a Green kernel estimate for convex sets of finite inradius. For a nonempty closed proper subset $K \subset M$, define

$$\operatorname{inrad}(K) = \sup_{p \in K} d_M(p, M \setminus K), \tag{3}$$

and set $\operatorname{inrad}(M) = \infty$. Thus finite inradius means that the radii of ambient balls contained in K are uniformly bounded. For $C > 0$, let $N_C(K)$ denote the tubular neighborhood $\{x \in M : d_M(x, K) < C\}$, and let G denote the positive minimal Green kernel of $-\Delta$.

Theorem 1.3. *Let $K \subset M$ be nonempty, closed, and convex. If $\operatorname{inrad}(K) < \infty$, then, for every $C > 0$,*

$$\sup_{x \in M} \int_{N_C(K)} G(x, y) \, \mathrm{d}\operatorname{vol}(y) < \infty. \tag{4}$$

The estimate (4) has a natural probabilistic interpretation: for Brownian motion with generator Δ started at x, the integral is the expected total time spent in $N_C(K)$. Its proof combines two complementary estimates. Finite inradius and convexity give a uniform bound on the expected time needed to leave a fixed neighborhood of K, while pinched negative curvature ensures that, after moving sufficiently far away from K, the probability of returning is uniformly bounded away from 1. Together, these estimates bound the expected total occupation time in $N_C(K)$ independently of the starting point.

To apply this result in the proof of Theorem 1.2, suppose first that $n \geq 3$, and let K be the convex hull of the limit set of $\iota(\mathbb{H}^2)$ in $\partial_\infty M$. This hull lies at finite Hausdorff distance from $\iota(\mathbb{H}^2)$, and $\bar{\iota} = \pi_K \circ \iota \colon \mathbb{H}^2 \to K$ is a quasi-isometry. A covering argument based on this quasi-isometry shows that K cannot contain arbitrarily large ambient balls. Thus K has finite inradius, and Theorem 1.3 gives (4) for this convex hull.

Using nearest point projection onto K and a coarse inverse of $\bar{\iota}$, we construct a smooth map $f \colon M \to \mathbb{H}^2$ such that $f \circ \iota$ is at uniformly bounded distance from $\mathrm{id}_{\mathbb{H}^2}$. The construction also ensures that the tension field of f decays exponentially with the distance to K. The Green kernel estimate (4), together with the geometric estimates away from K, allows us to apply the harmonic replacement argument of [12, Section 4, in particular Corollary 4.7] to obtain a harmonic map $h \colon M \to \mathbb{H}^2$ at uniformly bounded distance from f. Consequently, $h \circ \iota$ remains at uniformly bounded distance from $\mathrm{id}_{\mathbb{H}^2}$, since

$$d_{\mathbb{H}^2}\big(h(\iota(x)), x\big) \leq d_{\mathbb{H}^2}\big(h(\iota(x)), f(\iota(x))\big) + d_{\mathbb{H}^2}\big(f(\iota(x)), x\big).$$

This proves (2).

When $n = 2$, the embedding ι is itself a quasi-isometry. Applying the Benoist–Hulin theorem to a coarse inverse gives the desired harmonic map directly.

Organization of this paper. Section 2 studies the convex hull associated with ι and proves that it has finite inradius when $n \geq 3$. Section 3 records the projection and distance estimates needed for the Green kernel bound. Section 4 proves Theorem 1.3, and Section 5 completes the proof of Theorem 1.2.

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2. Convex hulls and finite inradius

Throughout the paper, M satisfies (1), and d_M denotes its Riemannian distance. A map $f\colon X \to Y$ is a (λ, c)-quasi-isometric embedding if $\lambda \geq 1$, $c \geq 0$, and

$$\lambda^{-1} d_X(x, x') - c \leq d_Y\big(f(x), f(x')\big) \leq \lambda d_X(x, x') + c$$

for all $x, x' \in X$. It is a quasi-isometry if, in addition, there exists $C \geq 0$ such that every $y \in Y$ satisfies

$$d_Y(y, f(x)) \leq C$$

for some $x \in X$; see [3, Definition I.8.14].

Fix a (λ, ε)-quasi-isometric embedding $\iota\colon \mathbb{H}^2 \to M$ (with respect to the ambient distance d_M). Let $\partial_\infty M$ denote the ideal boundary of M, consisting of equivalence classes of geodesic rays that remain at bounded distance from one another. The space $\overline{M} = M \cup \partial_\infty M$, equipped with the cone topology, is called the visual compactification of M. For $x, y \in \overline{M}$, denote by $[x, y]$ the generalized geodesic joining them, with $[x, x] = \{x\}$. Thus $[x, y]$ is a segment, a ray, or a complete geodesic, together with its ideal endpoints when present. A subset of $\overline{M}$ is convex if it contains $[x, y]$ whenever it contains x and y. For a closed set $A \subset \overline{M}$, let $\mathrm{CH}(A)$ be the intersection with M of its closed convex hull in $\overline{M}$, and put

$$\mathrm{Join}(A) = \left(\bigcup_{x,y \in A} [x, y] \right) \cap M.$$

Bowditch's lemma [2, Lemma 2.6] gives $\sigma = \sigma(a, b) < \infty$ such that

$$\mathrm{CH}(A) \subset N_\sigma(\mathrm{Join}(A)) \tag{5}$$

for every closed $A \subset \overline{M}$, where the tubular neighborhood here is defined via the Riemannian distance d_M.

The map ι induces a continuous injective boundary map $\partial_\infty \iota\colon S^1 \to \partial_\infty M$; see [3, Chapter III.H]. Set $Q = \iota(\mathbb{H}^2) \subset M$, $S = \partial_\infty \iota(S^1)$, and $K = \mathrm{CH}(S) \subset M$. The set S is compact, and K is nonempty, closed, and convex. We write d_{H} for the Hausdorff distance between subsets of M, measured with respect to the Riemannian distance d_M.

Although Q need not be convex, its geometry is controlled, up to bounded Hausdorff distance, by its ideal boundary S. The Morse lemma [3, Chapter III.H] gives the following comparison between Q and the geodesic join of its ideal boundary.

Lemma 2.1. *There exists $D_0 = D_0(a, \lambda, \varepsilon) < \infty$ such that*

$$d_{\mathrm{H}}(Q, \mathrm{Join}(S)) \leq D_0.$$

Let $\pi_K\colon M \to K$ be nearest point projection, which is 1-Lipschitz, and put $\bar{\iota} = \pi_K \circ \iota$.

Corollary 2.2. *With D_0 as in Lemma 2.1 and σ as in (5), one has*

$$d_{\mathrm{H}}(Q, K) \le D_0 + \sigma.$$

Moreover, the nearest point projection $\bar{\iota} = \pi_K \circ \iota\colon (\mathbb{H}^2, d_{\mathbb{H}^2}) \to (K, d_M)$ is a quasi-isometry, with constants depending only on a, b in (1), and the quasi-isometry constants of ι.

Proof. Since $\mathrm{Join}(S) \subset K$, Lemma 2.1 gives

$$d_M(\iota(x), \bar{\iota}(x)) = d_M(\iota(x), K) \le D_0$$

for every $x \in \mathbb{H}^2$. By (5),

$$K \subset N_\sigma(\mathrm{Join}(S)) \subset N_{\sigma+D_0}(Q). \tag{6}$$

Recall that ι is a (λ, ε)-quasi-isometric embedding. The triangle inequality and the 1-Lipschitz property of π_K give

$$\lambda^{-1} d_{\mathbb{H}^2}(x, y) - \varepsilon - 2D_0 \le d_M(\bar{\iota}(x), \bar{\iota}(y)) \le \lambda d_{\mathbb{H}^2}(x, y) + \varepsilon$$

for all $x, y \in \mathbb{H}^2$. Thus $\bar{\iota}$ is a quasi-isometric embedding.

Finally, for every $k \in K$, the inclusion (6) provides $x \in \mathbb{H}^2$ such that $d_M(k, \iota(x)) \le \sigma + D_0$. Since $\pi_K(k) = k$, we obtain

$$\begin{aligned} d_M(k, \bar{\iota}(x)) &= d_M(\pi_K(k), \pi_K(\iota(x))) \\ &\le d_M(k, \iota(x)) \le \sigma + D_0. \end{aligned}$$

Hence $\bar{\iota} : \mathbb{H}^2 \to K$ is a quasi-isometry. □

We next construct a coarse inverse of $\bar{\iota}$, and extend it smoothly to M. For a smooth map f, let Df and $D^2 f$ denote its first and second covariant derivatives.

Lemma 2.3. *There is a smooth map $\widehat{q}\colon M \to \mathbb{H}^2$ with*

$$\sup_{z \in M} \left(\|D\widehat{q}_z\| + \|D^2 \widehat{q}_z\| \right) < \infty \tag{7}$$

whose restriction $q = \widehat{q}|_K$ is a quasi-isometry and satisfies

$$E := \sup_{x \in \mathbb{H}^2} d_{\mathbb{H}^2}\left(q(\bar{\iota}(x)), x\right) < \infty. \tag{8}$$

Proof. From Corollary 2.2, $\bar{\iota}\colon \mathbb{H}^2 \to K$ is a quasi-isometry, then there exists a quasi-isometry $q_0\colon K \to \mathbb{H}^2$, such that

$$\sup_{x \in \mathbb{H}^2} d_{\mathbb{H}^2}(q_0(\bar{\iota}(x)), x) \le E_0.$$

for some constant $E_0 < \infty$. Moreover, the map $F = q_0 \circ \pi_K\colon M \to \mathbb{H}^2$ satisfies, for some $L_0, C_0 < \infty$,

$$d_{\mathbb{H}^2}(F(u), F(v)) \le L_0 d_M(u, v) + C_0$$

for all $u, v \in M$. By [1, Proposition 2.4], there is a smooth map $\widehat{q}\colon M \to \mathbb{H}^2$ satisfying (7) and

$$A := \sup_{z \in M} d_{\mathbb{H}^2}(\widehat{q}(z), F(z)) < \infty. \tag{9}$$

Consequently, set $q = \widehat{q}|_K$, then (9) reads

$$\sup_{k\in K} d_{\mathbb{H}^2}(q(k), q_0(k)) \le A.$$

Thus q is at bounded distance from the quasi-isometry q_0 and is therefore itself a quasi-isometry. Moreover, for every $x \in \mathbb{H}^2$, the triangle inequality gives

$$\begin{aligned} d_{\mathbb{H}^2}(q(\bar{\iota}(x)), x) &\le d_{\mathbb{H}^2}(q(\bar{\iota}(x)), q_0(\bar{\iota}(x))) \\ &\quad + d_{\mathbb{H}^2}(q_0(\bar{\iota}(x)), x) \\ &\le A + E_0. \end{aligned}$$

This proves (8). □

Recall the inradius of K is defined as (3), or

$$\operatorname{inrad}(K) := \sup\{r \ge 0 : B(p, r) \subset K \text{ for some } p \in K\},$$

where $B(p, r)$ denotes the geodesic ball in M. We use the continuous quasi-isometry $q\colon K \to \mathbb{H}^2$ to construct an open cover of K with uniformly bounded diameter and multiplicity at most three. The Lebesgue cubical covering theorem [9, Chapter IV] will then be used to show that these covering properties are incompatible with the existence of arbitrarily large geodesic balls in K.

Proposition 2.4. *If $n \ge 3$, then* $\operatorname{inrad}(K) < \infty$.

Proof. Let $\mathcal{U}$ be the cover of $\mathbb{H}^2$ by the open vertex stars of a regular locally finite geodesic triangulation. This cover has multiplicity at most three and mesh at most some $m < \infty$. Since $q : K \to \mathbb{H}^2$ is a continuous quasi-isometry, there exist $L \ge 1$ and $A_0 \ge 0$ such that

$$d(u, v) \le L\big(d_{\mathbb{H}^2}(q(u), q(v)) + A_0\big) \qquad (u, v \in K).$$

Thus the pullback cover $\mathcal{V} = \{q^{-1}(U) : U \in \mathcal{U}\}$ is open in K, has multiplicity at most three, and has mesh at most $T = L(m + A_0)$.

Suppose that $\operatorname{inrad}(K) = \infty$. Choose $p_j \in K$ and $R_j \to \infty$ such that $B(p_j, R_j) \subset K$. Identify $T_{p_j}M$ isometrically with $\mathbb{R}^n$, and set

$$\ell_j = \frac{R_j}{2\sqrt{n}}, \qquad Q_j = [-\ell_j, \ell_j]^n.$$

Then

$$\exp_{p_j}(Q_j) \subset \overline{B}(p_j, R_j/2) \subset K.$$

Pulling $\mathcal{V}$ back to Q_j via $\exp_{p_j}$ gives a relatively open cover of multiplicity at most three. Since M is a Hadamard manifold,

$$|v - w| \le d\big(\exp_{p_j} v, \exp_{p_j} w\big),$$

so this cover has Euclidean mesh at most T.

For sufficiently large j, the side length $2\ell_j$ of Q_j exceeds T. Hence no member of a finite subcover meets two opposite faces of Q_j. The Lebesgue cubical covering theorem [9, Chapter IV] then implies that the subcover has multiplicity at least $n + 1 \ge 4$, a contradiction. □

3. Projection and distance estimates

Throughout this section, let $K \subset M$ be a nonempty closed convex set. Write $r(x) = d(x, K)$, and denote by $\pi_K \colon M \to K$ the nearest-point projection. All constants below depend only on n, a, b, and are independent of K. For a smooth map f, we write $\tau(f) = \operatorname{tr} D^2 f$ for its tension field.

We first recall a smooth approximation result for π_K. Tošić proved that the local Lipschitz constant of π_K decays exponentially with the distance from K [12, Proposition 3.5]. Using the smoothing method of Benoist–Hulin [1, Section 2.2], he obtained the corresponding decay estimates for the first two derivatives of a smooth approximation [12, Corollary 3.7].

Lemma 3.1. *The nearest point projection π_K admits a smooth approximation $p \colon M \to M$ for which there are constants $D, C > 0$ such that*

$$d\big(p(x), \pi_K(x)\big) \le D, \tag{10}$$

$$\|Dp_x\| + \|D^2 p_x\| \le C e^{-a r(x)} \qquad (x \in M). \tag{11}$$

Set $\delta(x) = d(x, p(x))$. Since $r(x) = d(x, \pi_K(x))$, the reverse triangle inequality and (10) give

$$|\delta(x) - r(x)| \le D. \tag{12}$$

In particular, $\delta(x) > 0$ whenever $r(x) > D$, which means δ is smooth on $\{r > D\}$ and thus provides a smooth substitute for r outside a fixed neighborhood of K.

Tošić established a positive lower bound for $\Delta\delta$ and a uniform bound for $|\nabla\delta|$ sufficiently far from K [12, Proposition 4.1]. The same argument, using Hessian comparison, also gives a uniform bound for $|\Delta\delta|$. We record these estimates below.

Lemma 3.2. *There are constants $R_0, A, B > 0$ such that δ is smooth on $\{r > R_0\}$ and satisfies*

$$\Delta\delta \ge A, \qquad |\nabla\delta| \le B, \qquad |\Delta\delta| \le B \tag{13}$$

on this set.

4. Green Kernel Integral Bounds from Finite Inradius

We now prove Theorem 1.3. Let $H(t, x, y)$ denote the heat kernel associated with $\partial_t u = \Delta u$. McKean's spectral estimate [10] gives

$$\lambda_1(-\Delta) \ge \frac{(n-1)^2 a^2}{4} > 0.$$

Consequently,

$$G(x, y) = \int_0^\infty H(t, x, y)\, \mathrm{d}t \tag{14}$$

defines the positive minimal Green kernel of $-\Delta$ for $x \ne y$ [6, Chapter 13].

Let (X_t) be Brownian motion on M with generator Δ. For $x \in M$, denote by $\mathbb{P}_x$ and $\mathbb{E}_x$ its probability law and expectation, respectively, when started at x. Its transition density is then $H(t, x, y)$. Since $\mathrm{Ric}_M \ge -(n-1)b^2$, the manifold

is stochastically complete: Brownian motion has infinite lifetime almost surely, or equivalently,

$$\int_M H(t,x,y)\,\mathrm{d}\,\mathrm{vol}(y) = 1 \qquad (t>0,\ x\in M)$$

[8, Theorem 4.2.4].

For every Borel set $E \subset M$, Tonelli's theorem and (14) therefore give

$$\mathbb{E}_x\left(\int_0^\infty \mathbf{1}_E(X_t)\,\mathrm{d}t\right) = \int_E G(x,y)\,\mathrm{d}\,\mathrm{vol}(y). \tag{15}$$

We use the standard stopping-time and strong Markov properties of Brownian motion as presented in [8, Section 1.3].

Lemma 4.1. *For every $R>0$, there is $p_R = p_R(n,b,R) > 0$ such that*

$$\mathbb{P}_x\{X_1 \in B(y,1)\} \geq p_R$$

whenever $d(x,y) \leq R$.

Proof. By Cheeger–Yau comparison [4] and $\mathrm{Ric}_M \geq -(n-1)b^2$, there is $c_R > 0$ such that

$$H(1,x,z) \geq c_R \qquad \text{whenever } d(x,z) \leq R+1.$$

If $d(x,y) \leq R$, then $d(x,z) < R+1$ for every $z \in B(y,1)$. Since M is a Hadamard manifold, volume comparison gives $\mathrm{vol}\,B(y,1) \geq \omega_n$, where ω_n is the volume of the Euclidean unit ball. Hence

$$\begin{aligned}\mathbb{P}_x\{X_1 \in B(y,1)\} &= \int_{B(y,1)} H(1,x,z)\,\mathrm{d}\,\mathrm{vol}(z)\\ &\geq c_R\,\mathrm{vol}\,B(y,1) \geq c_R\omega_n.\end{aligned}$$

Taking $p_R = c_R\omega_n$ proves the assertion. □

For the remainder of this section, fix a nonempty closed convex set $K \subset M$ with $W = \mathrm{inrad}(K) < \infty$, and retain the notation $r = d(\cdot, K)$. By (15), to prove Theorem 1.3, it suffices to show that, for every $C > 0$, the expected total time Brownian motion spends in $N_C(K)$ is uniformly bounded over all starting points.

We begin by considering how long Brownian motion can remain in a fixed neighborhood of K before leaving it. The following lemma gives a uniform bound on this expected exit time.

Lemma 4.2. *For every $L > 0$, the first exit time*

$$\tau_L = \inf\{t \geq 0 : X_t \notin N_L(K)\}$$

satisfies

$$E_L := \sup_{z\in N_L(K)} \mathbb{E}_z\tau_L < \infty. \tag{16}$$

Proof. We first show that every point of $N_L(K)$ lies within a uniformly bounded distance of a unit ball outside this neighborhood.

Fix $z \in N_L(K)$ and put $z_0 = \pi_K(z)$. By (3), there is $u \notin K$ with $d(z_0,u) < W+1$. Set $u_0 = \pi_K(u)$. Since π_K is 1-Lipschitz, there holds $d(z_0,u_0) < W+1$. The characterization of nearest point projection implies that every point on the geodesic

ray from u_0 through u projects to u_0. Choose y on this ray with $d(y,u_0)=L+2$. Then $r(y)=L+2$, so $B(y,1)\subset M\setminus N_L(K)$, while

$$\begin{aligned} d(z,y) &\le d(z,z_0)+d(z_0,u_0)+d(u_0,y)\\ &< L+(W+1)+(L+2)=2L+W+3. \end{aligned}$$

Consequently, Lemma 4.1 gives

$$\mathbb{P}_z\{\tau_L\le 1\}\ge \mathbb{P}_z\{X_1\in B(y,1)\}\ge p_{2L+W+3}=:p_*>0.$$

This lower bound is independent of z.

Iterating the Markov property at integer times yields

$$\mathbb{P}_z\{\tau_L>k\}\le (1-p_*)^k \qquad (k=0,1,\ldots).$$

Hence

$$\mathbb{E}_z\tau_L\le \sum_{k=0}^{\infty}\mathbb{P}_z\{\tau_L>k\}\le \sum_{k=0}^{\infty}(1-p_*)^k=p_*^{-1}.$$

This proves (16). □

The uniform exit-time estimate (16) does not by itself control the expected total time Brownian motion spends near K, since the path may return after leaving the neighborhood. To control these returns, we next show that, once Brownian motion has moved sufficiently far from K, the probability of reaching a smaller neighborhood of K again is uniformly bounded away from one.

Lemma 4.3. *Let D be the constant in (12) and R_0 the constant in Lemma 3.2. There exists $\kappa>0$, depending only on n,a,b, such that the following holds. For any $C>R_0$, define*

$$\sigma_C=\inf\{t\ge 0: r(X_t)\le C\}.$$

Then, for every $L>C+2D$,

$$\sup_{\{z:\, r(z)\ge L\}}\mathbb{P}_z\{\sigma_C<\infty\}\le e^{-\kappa(L-C-2D)}<1. \tag{17}$$

Proof. Let A,B be the constants in Lemma 3.2. Set $\kappa=A/(2B^2)$ and define $w=e^{\kappa(C+D-\delta)}$ on $\{r>R_0\}$. By Lemma 3.2,

$$\begin{aligned} \Delta w &= w\big(\kappa^2|\nabla\delta|^2-\kappa\Delta\delta\big)\\ &\le (\kappa^2B^2-\kappa A)w=-\frac{A^2}{4B^2}w\le 0. \end{aligned}$$

Since $|\delta-r|\le D$ by (12), we have $0<w\le e^{2\kappa D}$ on $\{r\ge C\}$ and $w\ge 1$ on $\{r=C\}$. Moreover, for every z with $r(z)\ge L$,

$$w(z)\le e^{-\kappa(r(z)-C-2D)}\le e^{-\kappa(L-C-2D)}.$$

Fix such a point z. By Itô's formula and stochastic completeness, $w(X_{t\wedge\sigma_C})$ is a bounded supermartingale under $\mathbb{P}_z$. Since $r(X_{\sigma_C})=C$ on $\{\sigma_C<\infty\}$, it follows that

$$\begin{aligned} \mathbb{P}_z\{\sigma_C\le t\} &\le \mathbb{E}_z[w(X_{t\wedge\sigma_C})]\\ &\le w(z)\le e^{-\kappa(L-C-2D)}. \end{aligned}$$

Letting $t\to\infty$ and taking the supremum over all z with $r(z)\ge L$ proves (17). □

The estimate for exit-time (16) controls the expected duration of a visit near K, while the estimate for the probability of return (17) controls repeated visits. Combining these estimates, we establish the uniform Green kernel bound (4).

Proof of Theorem 1.3. It suffices to prove (4) for $C > R_0$. Fix such a C, choose $L > C + 2D$, and write $\theta = e^{-\kappa(L-C-2D)} < 1$.

Fix a starting point $x \in M$. Let $\alpha_1 = \sigma_C$ be the first hitting time of $\{r \le C\}$, and let β_1 be the first exit time from $N_L(K)$ after α_1. Recursively, let α_{j+1} be the first hitting time of $\{r \le C\}$ after β_j, and let β_{j+1} be the subsequent exit time from $N_L(K)$. If any of these times is infinite, set all subsequent times equal to infinity.

We first estimate the expected duration of each interval $[\alpha_j, \beta_j]$. By Lemma 4.2 and the strong Markov property, $\beta_j < \infty$ almost surely on $\{\alpha_j < \infty\}$. Define $T_j = \beta_j - \alpha_j$ on this event and $T_j = 0$ otherwise. Applying the strong Markov property at α_j gives

$$\mathbb{E}_x T_j \le E_L \mathbb{P}_x\{\alpha_j < \infty\}. \tag{18}$$

We next estimate the probability of successive returns. At every finite β_j, path continuity gives $r(X_{\beta_j}) = L$. Thus Lemma 4.3 and the strong Markov property at β_j imply

$$\mathbb{P}_x\{\alpha_{j+1} < \infty\} \le \theta \mathbb{P}_x\{\beta_j < \infty\} = \theta \mathbb{P}_x\{\alpha_j < \infty\}.$$

Since $\mathbb{P}_x\{\alpha_1 < \infty\} \le 1$, induction yields

$$\mathbb{P}_x\{\alpha_j < \infty\} \le \theta^{j-1}. \tag{19}$$

Since $L > C$ and $r(X_t)$ is continuous, successive crossings between $\{r = C\}$ and $\{r = L\}$ cannot accumulate in finite time. Thus, every visit to $N_C(K)$ is contained in one of the intervals $[\alpha_j, \beta_j]$. The total time spent in $N_C(K)$ is therefore bounded above by the sum of the lengths of these intervals. Hence, by (18)–(19) and Tonelli's theorem,

$$\begin{aligned}\mathbb{E}_x\left(\int_0^\infty \mathbf{1}_{N_C(K)}(X_t)\, \mathrm{d}t\right) &\le \sum_{j=1}^\infty \mathbb{E}_x T_j \\ &\le E_L \sum_{j=1}^\infty \theta^{j-1} = \frac{E_L}{1-\theta}.\end{aligned}$$

The bound is independent of x, so (15) gives (4). □

5. Harmonic replacement and the main theorem

We now turn to the construction of the harmonic map in Theorem 1.2. As a preparation for this construction, we require two technical propositions. In these propositions, let $K \subset M$ be a nonempty closed convex set with $\operatorname{inrad}(K) < \infty$, and set $r = d(\cdot, K)$.

The estimates in Lemma 3.2 and (4) allow us to apply the argument of [12, Lemma 4.6], yielding the following,

Proposition 5.1. *There exists a bounded function $\Phi\colon M \to [0, \infty)$ such that*

$$\Delta\Phi \ge e^{-ar} \tag{20}$$

in the distributional sense.

Using this function, the argument of [12, Corollary 4.7] gives the following existence result for harmonic maps.

Proposition 5.2. *Let N be a pinched Hadamard manifold and $f\colon M \to N$ a smooth map satisfying*

$$\|\tau(f)(x)\| \le T_f e^{-ar(x)} \qquad (x \in M) \tag{21}$$

for some constant $T_f < \infty$. Then there exists a harmonic map $h\colon M \to N$ such that

$$\sup_{x\in M} d_N\big(h(x), f(x)\big) < \infty.$$

We now complete the proof of Theorem 1.2. Recall that $S = \partial_\infty \iota(S^1)$, $K = \mathrm{CH}(S)$, and $r = d(\cdot, K)$. Let $\pi_K\colon M \to K$ be the nearest point projection and $\bar\iota = \pi_K \circ \iota$. We also recall the smooth approximation $p\colon M \to M$ of π_K from Lemma 3.1, and the smooth extension $\widehat q\colon M \to \mathbb{H}^2$ of the coarse inverse $q\colon K \to \mathbb{H}^2$ from Lemma 2.3.

Proof of Theorem 1.2. Suppose first that $n \ge 3$. By Proposition 2.4, K has finite inradius. Hence Theorem 1.3 gives (4). Let p be the map in Lemma 3.1, and set $f = \widehat q \circ p$. The covariant chain rule gives

$$D^2 f(u,v) = D^2\widehat q\big(Dp(u), Dp(v)\big) + D\widehat q\big(D^2 p(u,v)\big).$$

By (7) and (11),

$$\|Df_x\| + \|D^2 f_x\| \le C\big(e^{-ar(x)} + e^{-2ar(x)}\big) \le C' e^{-ar(x)}.$$

Taking the trace gives (21). Thus Proposition 5.2 supplies a harmonic map $h\colon M \to \mathbb{H}^2$ with

$$H_0 := \sup_{z\in M} d_{\mathbb{H}^2}(h(z), f(z)) < \infty. \tag{22}$$

Set $L_q = \sup_M \|D\widehat q\| < \infty$. Equation (10) gives

$$d_{\mathbb{H}^2}\big(f(\iota(x)), q(\bar\iota(x))\big) \le L_q\, d_M\big(p(\iota(x)), \pi_K(\iota(x))\big) \le L_q D.$$

Combining this with bounds (8) and (22), we obtain

$$\begin{aligned} d_{\mathbb{H}^2}\big(h(\iota(x)), x\big) &\le d_{\mathbb{H}^2}\big(h(\iota(x)), f(\iota(x))\big) \\ &\quad + d_{\mathbb{H}^2}\big(f(\iota(x)), q(\bar\iota(x))\big) \\ &\quad + d_{\mathbb{H}^2}\big(q(\bar\iota(x)), x\big) \\ &\le H_0 + L_q D + E. \end{aligned}$$

This proves (2) for $n \ge 3$.

It remains to consider $n = 2$. Since $\partial_\infty M \cong S^1$, the continuous injective boundary map $\partial_\infty \iota\colon S^1 \to \partial_\infty M$ is a homeomorphism. Thus $S = \partial_\infty M$ and $K = M$. By Corollary 2.2, ι is a quasi-isometry, and Lemma 2.3 provides a quasi-isometry $j\colon M \to \mathbb{H}^2$ satisfying

$$\sup_{x\in \mathbb{H}^2} d_{\mathbb{H}^2}\big(j(\iota(x)), x\big) < \infty.$$

By [1, Theorem 1.1], there exists a harmonic map $h\colon M \to \mathbb{H}^2$ satisfying

$$\sup_{z\in M} d_{\mathbb{H}^2}(h(z), j(z)) < \infty.$$

Consequently, the triangle inequality yields (2). □

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